



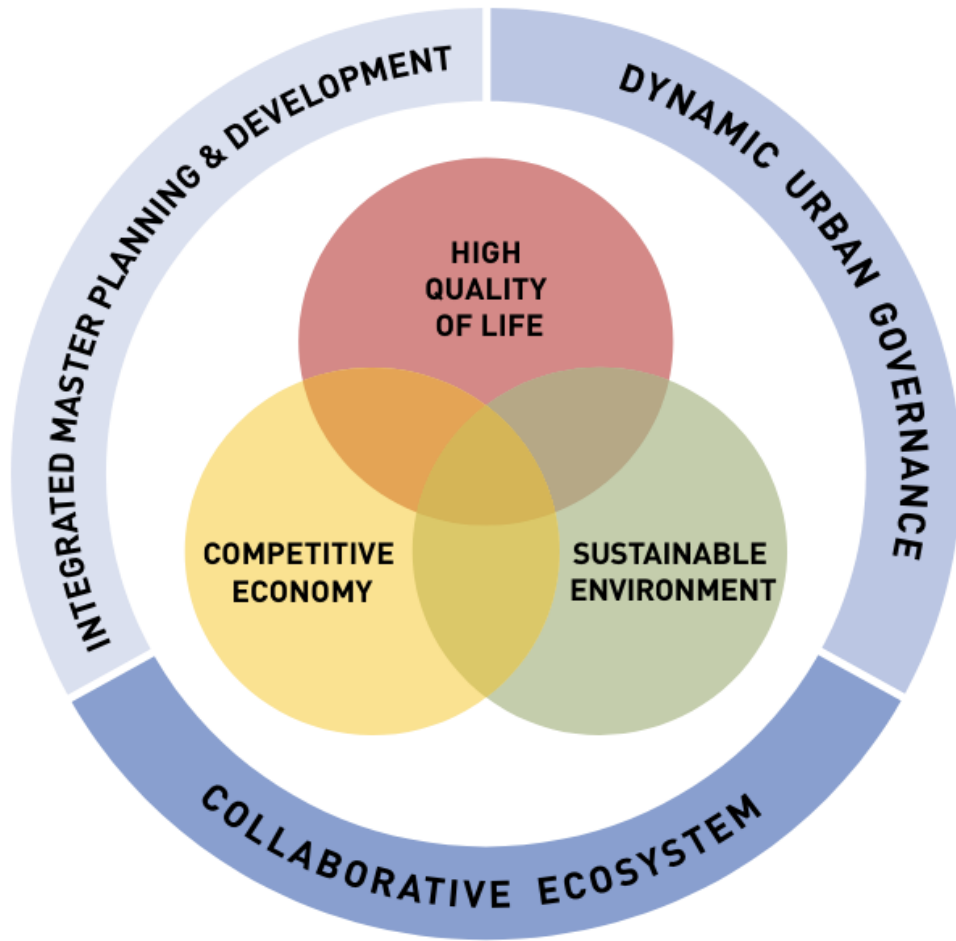
WELCOME TO CDE2501 Liveable Cities

SEMINAR 4
7 MARCH 2025

Lineup for Seminar 4

10.00am – 10.10am	Topic Introduction: “Smart Cities: Creating a Competitive Economy”
10.10am – 10.30am	Sharing by Mr Lim Chin Chong , Former Director of Specialised Parks Development Group, JTC
10.30am – 10.50am	Sharing by Adj Prof Koh Choon Fah , Adjunct Professor, NUS Cities; Former CEO of Edmund Tie & Company
10.50am – 11.00am	Break
11.00am – 11.10am	Synthesis
11.10am – 11.35am	Panel Discussion [Q&A]
11.35am – 11.40am	Closing & Admin Reminders to students

Course Learning Objectives



1. Understand how the urban systems approach has helped Singapore achieve its liveability outcomes.
2. Appreciate your role as future professionals to solve urban problems.

Image: The Singapore Liveability Framework 2024, Centre for Liveable Cities (2024), “Building Liveable & Sustainable Cities – A Framework for the Future”

Creating a Competitive Urban Economy

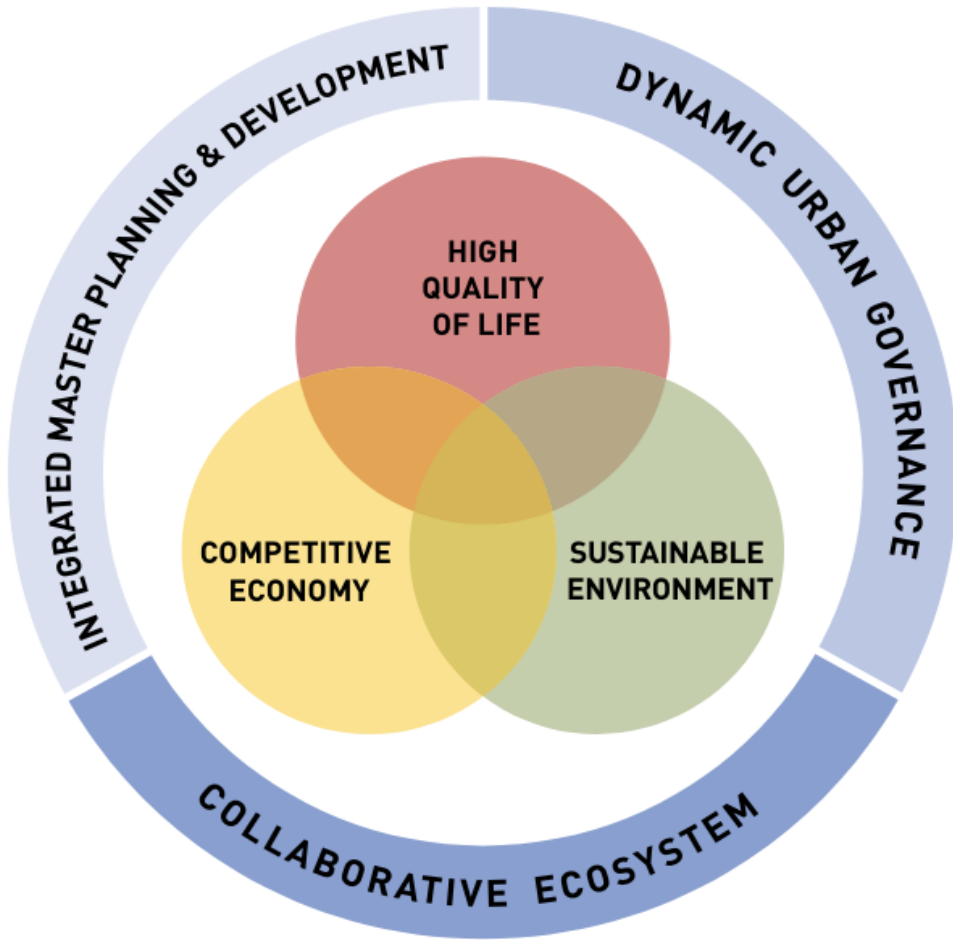


Image: The Singapore Liveability Framework 2024, Centre for Liveable Cities (2024), “Building Liveable & Sustainable Cities – A Framework for the Future”

Key Objectives of Seminar 4:

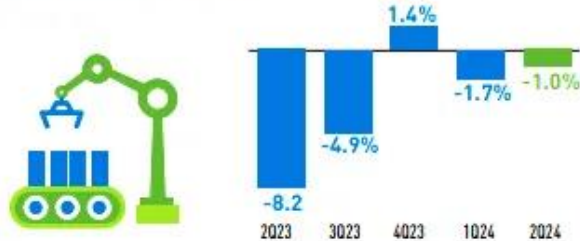
- Understand what contributes to a **competitive urban economy**
- Understand **why it is important** for cities to build competitive economies and how they can do it **amidst a changing global economy**
- Understand how **different industries and sectors** are important to Singapore’s competitiveness and **JTC’s and EDB’s strategies** to facilitate this amidst a changing domestic and global context
- Understand how **real estate** contributes to a **competitive economy** and how it achieves co-benefits for a **high quality of life** and **sustainable environment** as well

Remember!

The 3 outcomes **overlap** and are **interdependent** in achieving a liveable city. It represents the potential for synergies and the necessary trade-offs between the outcomes.

MAIN INDICATORS OF THE SINGAPORE ECONOMY

MANUFACTURING (YoY Growth)



CLUSTERS IN MANUFACTURING SECTOR



Source: Economic Survey
of Singapore 2023, MTI

OVERALL ECONOMY



GDP
at Current
Market Price

2022
\$687.2 billion
2023
\$673.3 billion



Real GDP
(YoY Growth)

2022
+3.8%
2023
+1.1%



GNI
Per Capita

2022
\$98,550
2023
\$95,455

STRUCTURE OF THE ECONOMY IN 2023 (% OF NOMINAL VA)



Services Producing
Industries
72.7%

Ownership
of Dwellings
3.6%

Goods Producing
Industries
23.6%

Manufacturing
18.6%

Construction
3.5%

BREAKDOWN OF SERVICES PRODUCING INDUSTRIES



Wholesale
Trade
22.3%



Professional
Services
5.7%



Administrative &
Support Services
2.8%



Accommodation
0.8%



Finance &
Insurance
13.8%



Information &
Communications
5.7%



Retail
Trade
1.3%



Other Services
Industries
9.6%



Transportation
& Storage
6.8%



Real
Estate
3.1%



Food & Beverage
Services
0.9%



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BROAD AND DETAILED STRUCTURE

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DETAILED STRUCTURE AND EXPLANATORY NOTES

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How would you define
the word “industry”?



Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



industry

/ˈɪndəstri/

noun

1. economic activity concerned with the processing of raw materials and manufacture of goods in factories.

"new investment incentives for British industry"

Similar:

manufacturing

production

fabrication

construction



2. hard work.

"the kitchen became a **hive of industry**"

Similar:

hard work

industriousness

diligence

assiduity

application





Source: Smokestacks in Manchester England c. 1858 watercolor by William Wyld



Source: Shutterstock



Source: Freepik, AI-generated

Three-sector model

22 languages

Article Talk

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From Wikipedia, the free encyclopedia



This article includes a list of general [references](#), but **it lacks sufficient corresponding inline citations**. Please help to [improve](#) this article by [introducing](#) more precise citations.
(August 2008) ([Learn how and when to remove this message](#))

The **three-sector model** in [economics](#) divides [economies](#) into three [sectors](#) of activity: extraction of [raw materials](#) ([primary](#)), [manufacturing](#) ([secondary](#)), and [service](#) industries which exist to facilitate the transport, distribution and sale of goods produced in the secondary sector ([tertiary](#)).^[1] The model was developed by [Allan Fisher](#),^{[2][3][4]} [Colin Clark](#),^[5] and [Jean Fourastié](#)^[6] in the first half of the 20th century, and is a representation of an [industrial economy](#). It has been criticised as inappropriate as a representation of the economy in the 21st century.^[7]

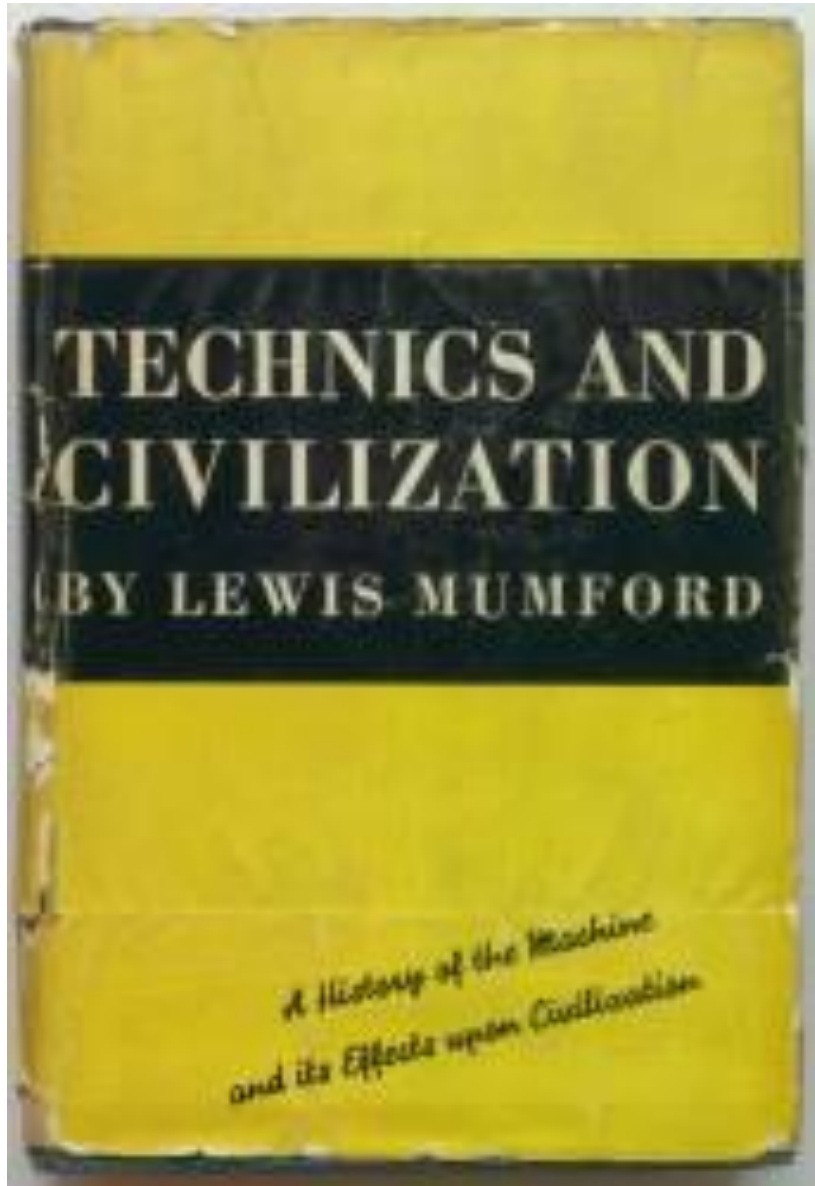
Economic sectors

Three-sector model

[Primary sector](#) (raw materials) ·
[Secondary sector](#) (manufacturing) ·
[Tertiary sector](#) (services)

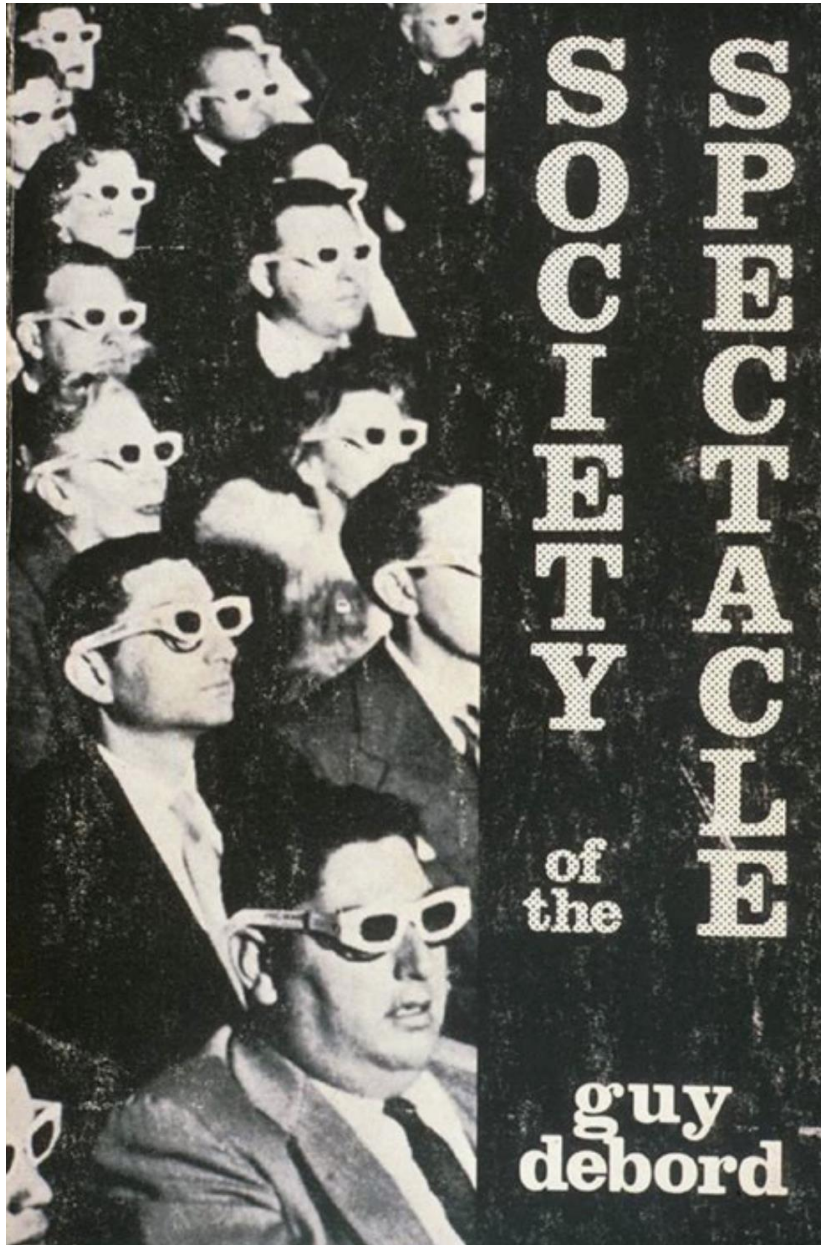
Additional sectors

[Quaternary sector](#) (information services) ·
[Quinary sector](#) (human services)



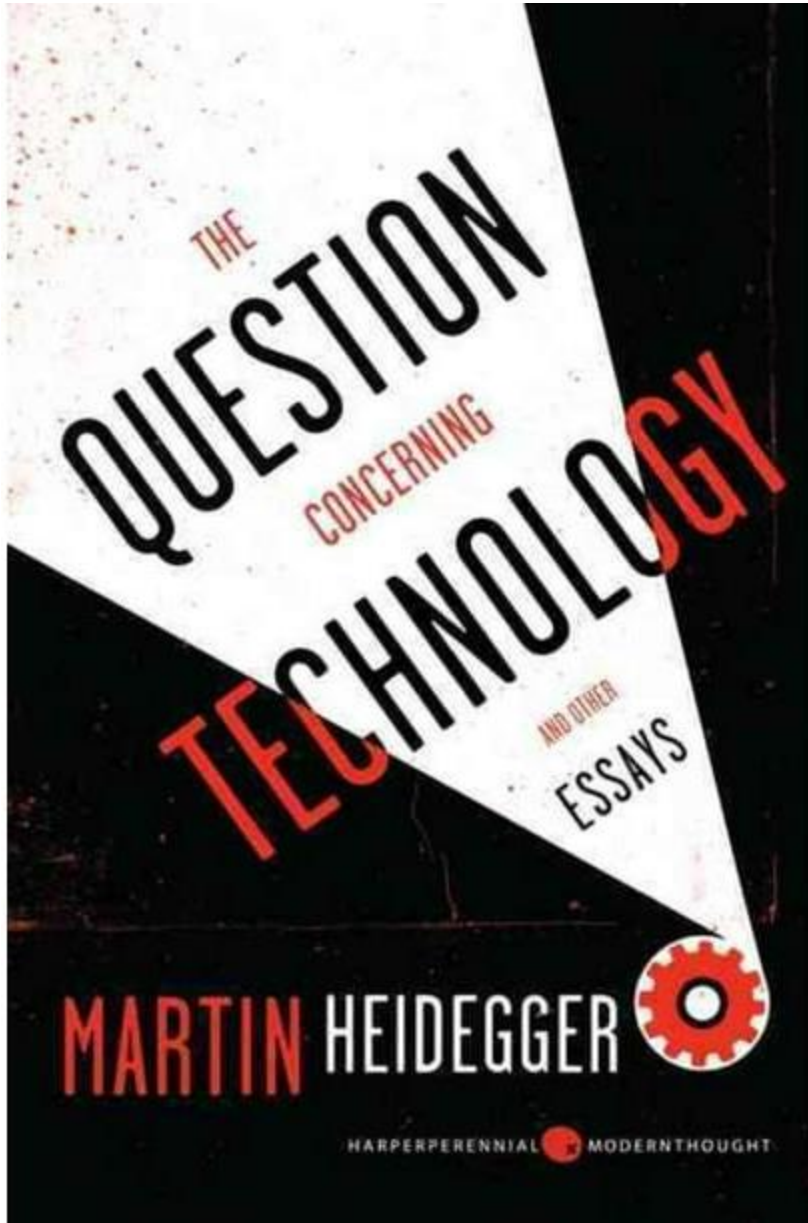
Cultural Preparation (1934): Monastery & the Clock

“The clock, not the steam-engine, is the key-machine of the modern industrial age. For every phase of its development the clock is both the outstanding fact and the typical symbol of the machine: even today no other machine is so ubiquitous. Here, at the very beginning of modern technics, appeared prophetically the accurate automatic machine which, only after centuries of further effort, was also to prove the final consummation of this technics in every department of industrial activity. [...] Now, the orderly punctual life that first took shape in the monasteries is not native to mankind, although by now Western peoples are so thoroughly regimented by the clock that it is ‘second nature’ and they look upon its observance as a fact of nature.”



Separation Perfected (1967)

“Understood in its totality, the spectacle is both the result and the project of the present mode of production. It is not a mere supplement or decoration added to the real world, it is the heart of this real society’s unreality. In all of its particular manifestations – news, propaganda, advertising, entertainment –the spectacle is the *model* of the prevailing way of life. It is the omnipresent affirmation of the choices that have *already been made* in the sphere of production and in the consumption implied by that production. In both form and content the spectacle serves as a total justification of the conditions and goals of the existing system. The spectacle is also the *constant presence* of this justification since it monopolizes the majority of the time spent outside the modern production process.”



“When Heidegger says that technology reveals things to us as ‘standing reserve,’ he means that everything is imposed upon or ‘challenged’ to be an orderly resource for technical application, which in turn we take as a resource for further use, and so on interminably. For example, we challenge land to yield coal, treating the land as nothing but a coal reserve. The coal is then stored, ‘on call, ready to deliver the sun’s warmth that is stored in it,’ which is then ‘challenged forth for heat, which in turn is ordered to deliver steam whose pressure turns the wheels that keep a factory running.’ The factories are themselves challenged to produce tools ‘through which once again machines are set to work and maintained.’”

Mark Blitz, *The New Atlantis*



Mr Lim Chin Chong

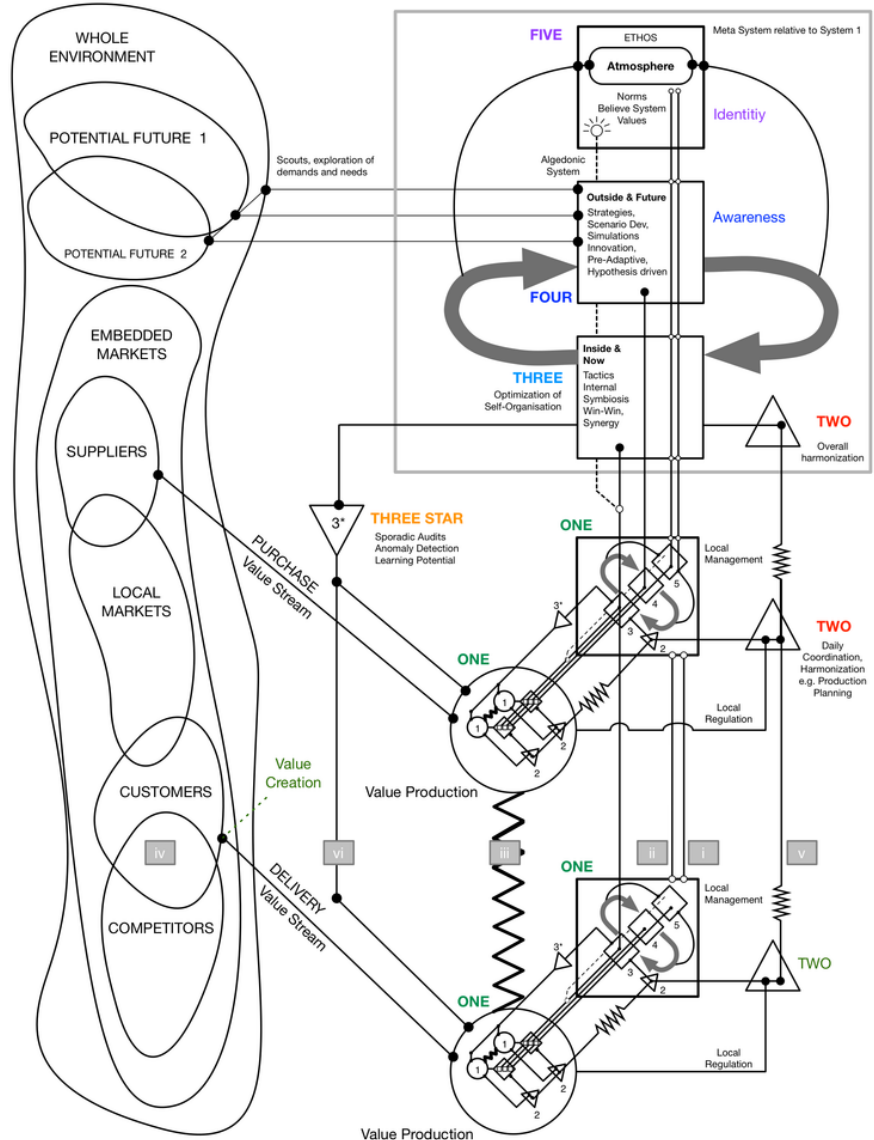
Former Director of Specialised Parks
Development Group, JTC



Adj Prof Koh Choon Fah

Adjunct Professor, NUS Cities

Former CEO of Edmund Tie & Company



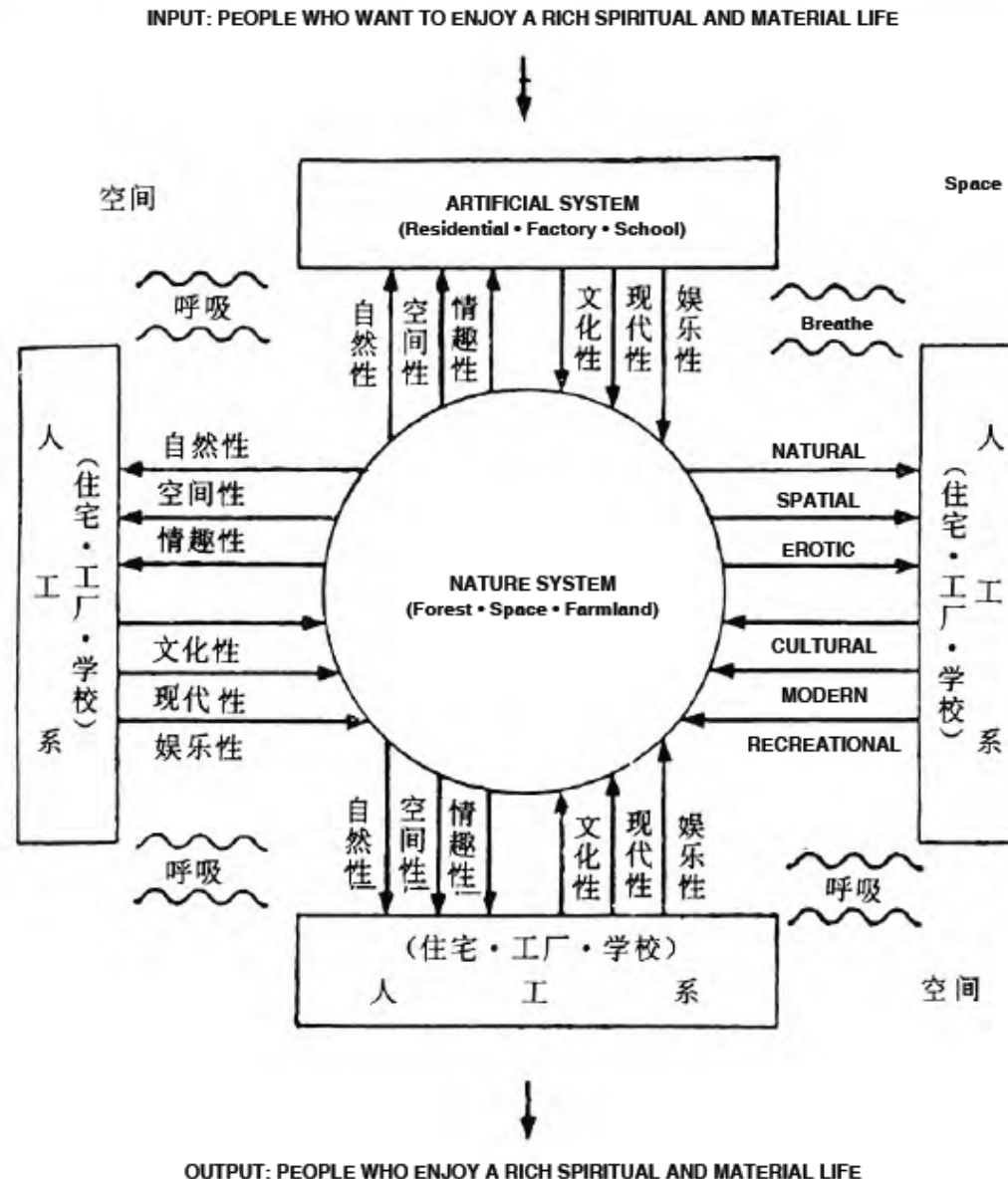
Viable System Model
Stafford Beer

Key Takeaways

1. We are duty-bound to control and maximize the value out of our built environment, but we need to strike a balance between the use of active technologies alongside passive and sustainable techniques.

Viable Systems Model by Stafford Beer

The diagram shows the three main elements: the operation, the management and the environment. The various arrows represent the many and often highly complex interactions between the various systems and the environment.



Key Takeaways

2. Industry and the harnessing of human and natural resources are instincts in humanity for its flourishing and survival. With a renewed awareness of how exploitative humanity can be, we have to collectively find a balance in advancing our economy but also safeguarding our society and natural resources.

Panel Discussion (Q&A)

Nobody has responded yet.

Hang tight! Responses are coming in.

Announcements and Reminders

Quiz #2: Competitive Economy!

- Please refresh your Canvas page:
 - Navigation Bar > Quizzes > Quiz 2: Competitive Economy
- Quiz is open until 11:59 PM tonight
- You have **8 minutes** to answer **5 multiple choice questions**.
- Open internet and open book

Announcements and Reminders

Next week: Studio 4 (Consultation)